

We commenced a Review of heat and cold stress in *Bos taurus* cattle from southern Australia during long-haul export by sea in April 2020. This is part of our commitment to improve animal welfare during export. It aligns with our responsibilities as a regulator. This is the first in a series of projects we will undertake to review cattle and buffalo exports by sea.

Issues relating to temperature stress in cattle and buffalo exports by sea were identified in the 2018 Technical Advisory Committee review of the Australian Standards for the Export of Livestock (ASEL). Some Independent Observer (IO) [voyage reports](#) and stakeholders raised similar issues.

About the review

The work undertaken for this review included:

- voyage data analysis
- an epidemiological analysis
- IO voyage report analysis
- review of Bureau of Meteorology data
- a literature review
- engagement with stakeholders in targeted stakeholder meetings and public consultation
- working with an independent technical expert group.

A draft report, setting out key findings and recommendations, was released for public consultation from 29 October 2021 to 3 December 2021. We received 11 submissions, and all submissions were considered in detail in finalising the review.

Outcome of the review

The review found the regulatory settings for the export of *Bos taurus* cattle by sea are effective in managing animal welfare. The review does not recommend changes to the current regulatory requirements. While animal welfare issues were identified in some voyage reports examined in the review, changes to standards (such as stocking density decreases and new bedding requirements) introduced under the revised ASEL on 1 November 2020 have addressed many of these.

We will use the report's findings to engage with industry to enhance the welfare of cattle in this supply chain.

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[Final Report Heat and Cold stress in *Bos taurus* \(PDF 3.11 MB\)](#) [Final Report Heat and Cold stress in *Bos taurus* \(DOCX 4.93 MB\)](#)

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About the technical expert group

We appointed an independent technical expert group. They reviewed the draft report and the final draft report and provided transparent feedback. This ensured we considered all issues when preparing the final report. We included their relevant feedback in the final report.

The expert group consisted of:

- Professor Andrew Fisher, Director, Animal Welfare Science Centre, University of Melbourne
- Associate Professor Anne Barnes, School of Veterinary Medicine, Murdoch University
- Dr Hugh Millar, Animal Health and Welfare Expert.

Source: <https://www.agriculture.gov.au/biosecurity-trade/export/controlled-goods/live-animals/animal-welfare/livestock-welfare-reviews/bos-taurus-review>